POLITECNICO DI TORINO

Master's Degree in Electronics

Master's Degree Thesis

Loss analysis of a 4-level,flying capacitor,bridge-less,totem-pole PFC

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MONTH 2026

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Chapter 1

SPICE model Validation

In order to properly analyze more critical components in terms of losses,it is useful to first create a digital model of the device;a validation procedure of the SPICE models of each device has been performed. The following sections describe the procedure and results of this validation procedure.

All simulation results are available [1]

1.1 Fast Leg GaN FET model validation

This section describes the behavior of the SPICE model of EPC2304, compared to the EPC datasheet [2]

1.1.1 Static charateristics

Output I-V charateristics

```
.temp 25

.dc Vds 0 6 0.1 Vgs list 2 3 4 5
```

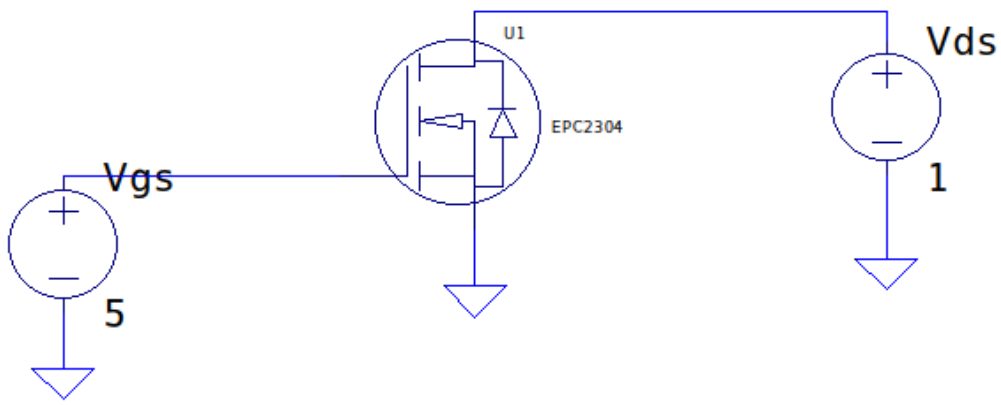

Figure 1.1: Schematic for I-V validation

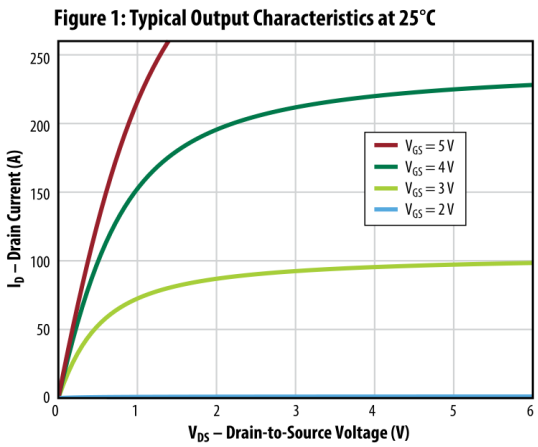

Figure 1.2: Datasheet I-V

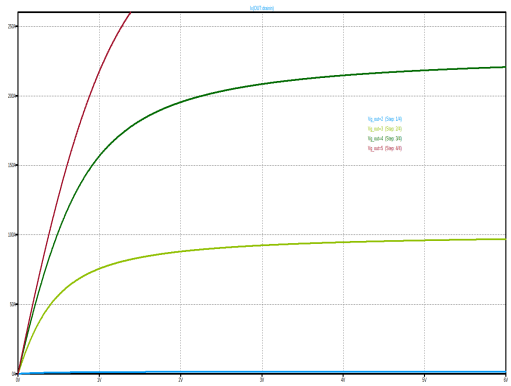

Figure 1.3: SPICE I-V

Vds@Vgs	Ids (datasheet)	Ids (simulation)
Vds=1V;Vgs=2V	$\approx 0A$	1.24A
Vds=6V;Vgs=3V	$\approx 100A$	97.47A
Vds=1V;Vgs=4V	$\approx 150A$	157.60A
Vds=1V;Vgs=5V	$\approx 215A$	218.568A

Table 1.1: I-V data comparision

Trans charateristics

```
.step temp 25 125 100

.temp 25

.dc Vgs 0.5 5 0.1
```

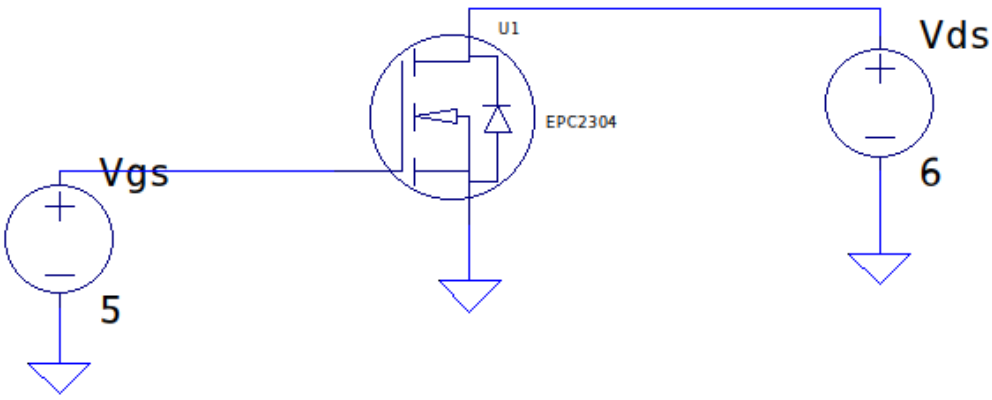

Figure 1.4: Schematic for trans charateristics validation

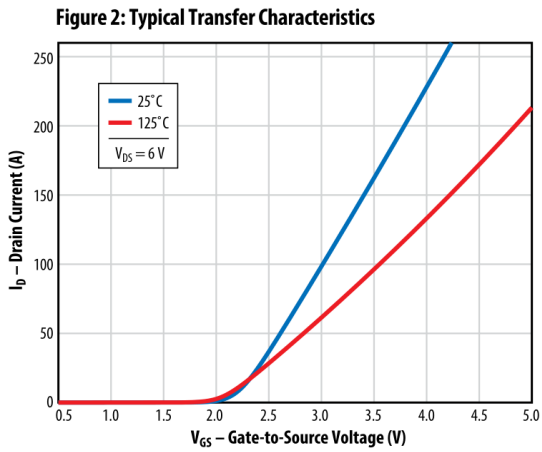

Figure 1.5: Datasheet trans charateristics

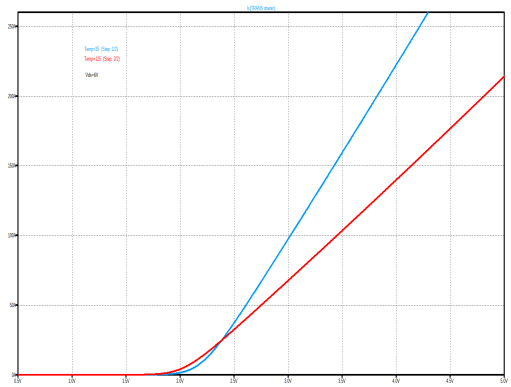

Figure 1.6: SPICE trans charateristics

Comparing the 2 it seems like the real device has a Vth of $\approx 2V$,while the simulated one has a Vth of $\approx 2.1V$

Vgs@T	Ids (datasheet)	Ids (simulation)
Vgs=2.5V;T=25°C	$\approx 40A$	37.396A
Vgs=3V;T=25°C	$\approx 100A$	97.4635A
Vgs=3.5;T=125°C	$\approx 100A$	103.57A
Vgs=4.5V;T=125°C	$\approx 175A$	176.82A

Table 1.2: I-V data comparision

Rdson

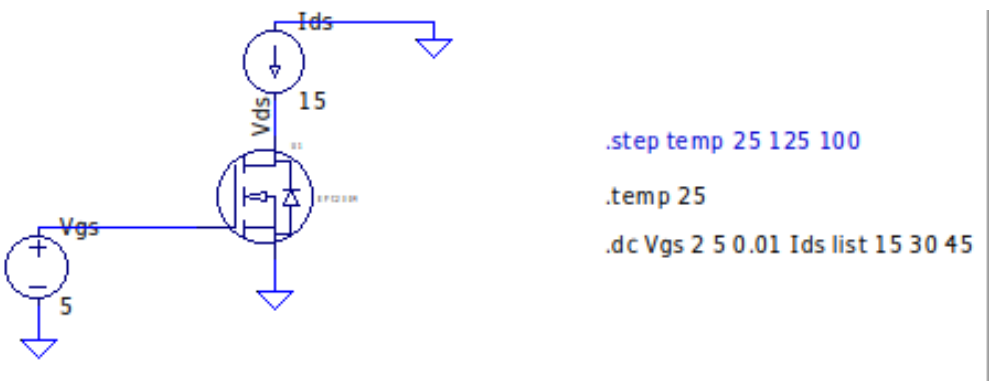

Figure 1.7: Schematic for Rdson validation

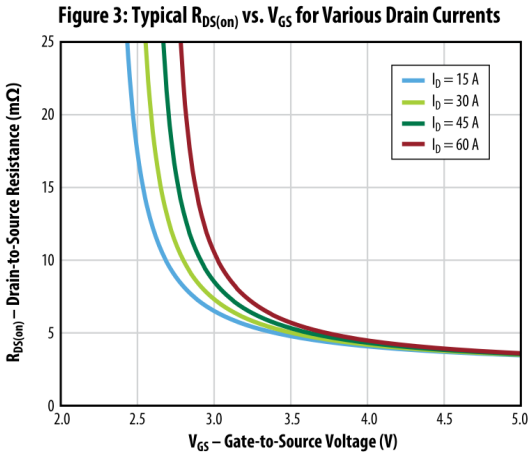

Figure 1.8: Datasheet Rdson

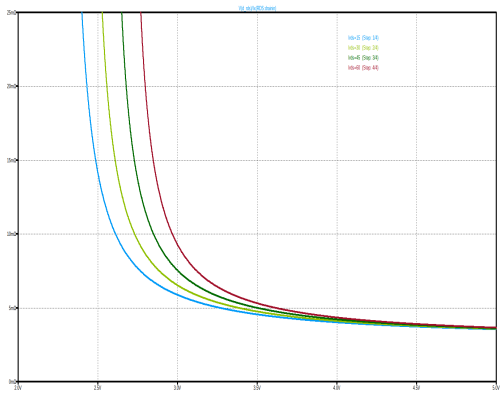

Figure 1.9: SPICE Rdson @25°C

Vgs@Ids	Rds (datasheet)	Rds (simulation)
Vgs=2.5V;Id=15A	$\approx 17.5m\Omega$	$14m\Omega$
Vgs=3V;Ids=60A	$\approx 10m\Omega$	$9.15m\Omega$
Vgs=3.5V;Ids=30A	$5m\Omega$	$4.73m\Omega$
Vgs=5V	$\approx 4m\Omega$	$3.64m\Omega$

Table 1.3: Rds data comparision

Rdson vs tempearture

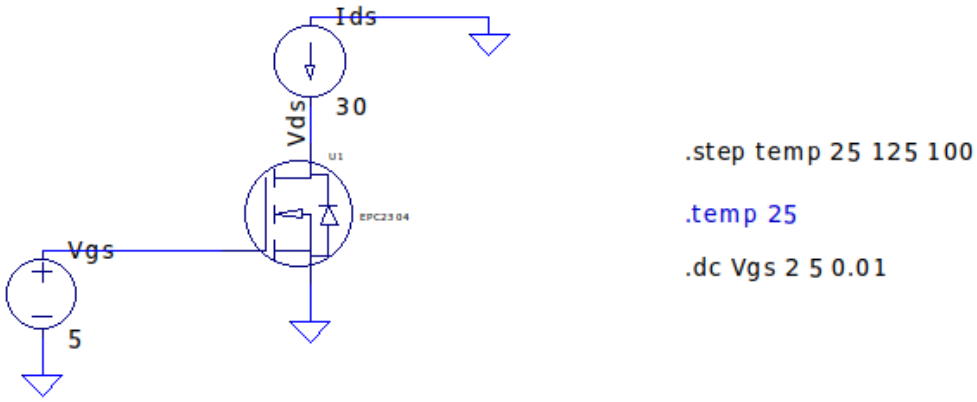

Figure 1.10: Schematic for Rdson vs tempearture

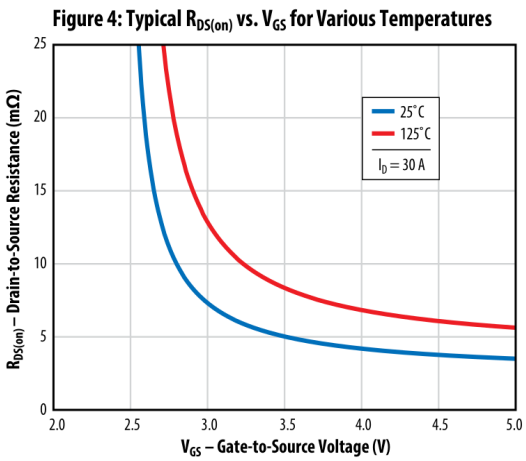

Figure 1.11: Datasheet Rdson vs tempearture

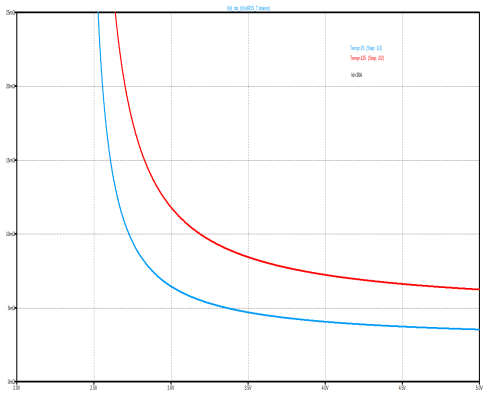

Figure 1.12: SPICE Rdson vs tempearture

Vgs@T	Rds (datasheet)	Rds (simulation)
Vgs=3.5V;T=25°C	$\approx 5m\Omega$	$4.69m\Omega$
Vgs=5V;T=125°C	$\approx 5.5m\Omega$	$6.29m\Omega$
Vgs=2.95V;T=25°C	$\approx 7m\Omega$	$6.82m\Omega$

Table 1.4: Rds temperature comparision

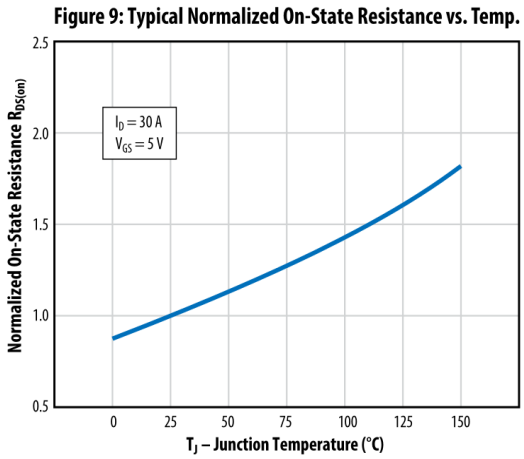

Figure 1.13: Datasheet Rdson vs temperature

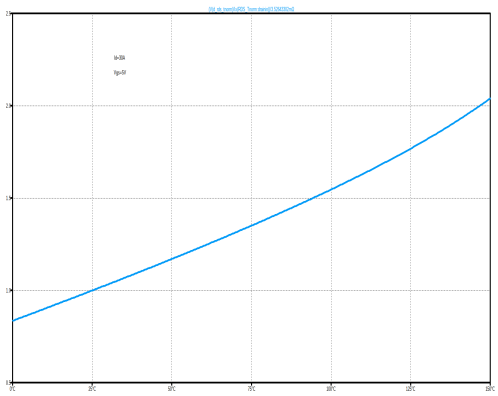

Figure 1.14: SPICE Rdson vs temperature

Reverse body diode

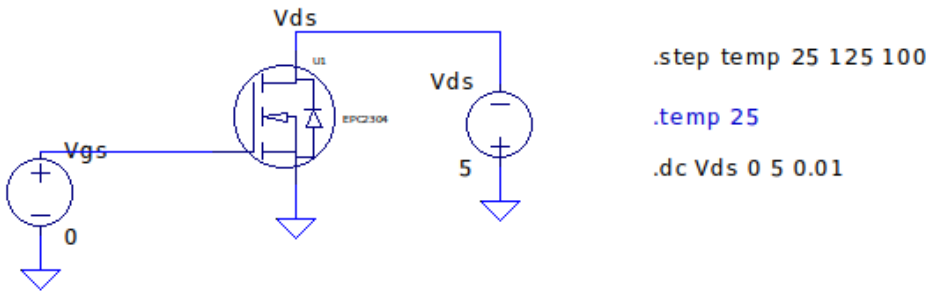

Figure 1.15: Schematic for Reverse body diode

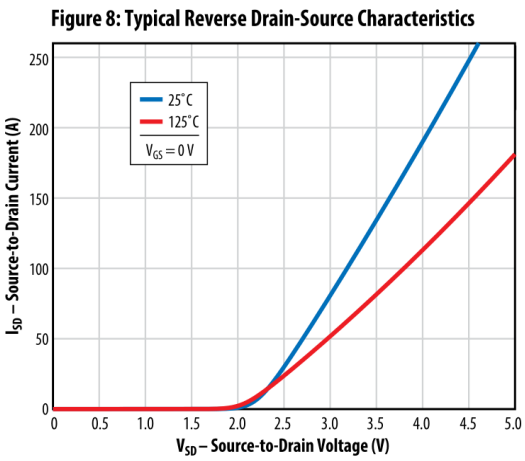

Figure 1.16: Datasheet Reverse body diode

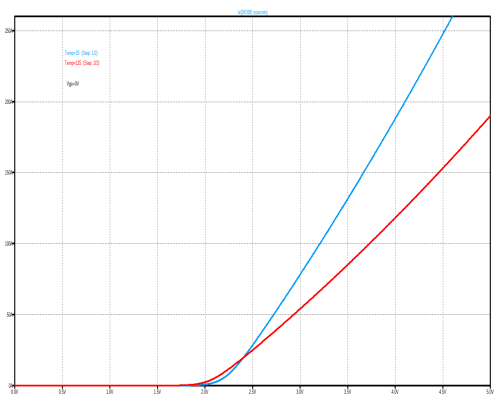

Figure 1.17: SPICE Reverse body diode

Vgs@T	Rds (datasheet)	Rds (simulation)
Vsd=4.5V;T=25°C	$\approx 250A$	$247.5A$
Vsd=3V;T=125°C	$\approx 50A$	54.21
Vsd=4.5V;T=125°C	$\approx 148A$	$153.22A$

Table 1.5: Reverse diode data comparision

Vth vs temperature

Figure 1.18: Schematic for Vth vs temperature

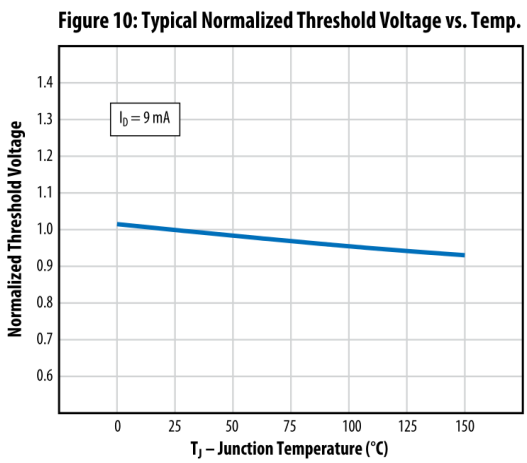

Figure 1.19: Datasheet Vth vs temperature

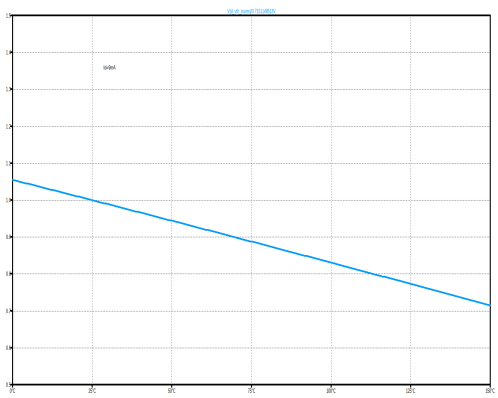

Figure 1.20: SPICE Vth vs temperature

1.1.2 Dynamic Charateristics

Typical switching curves

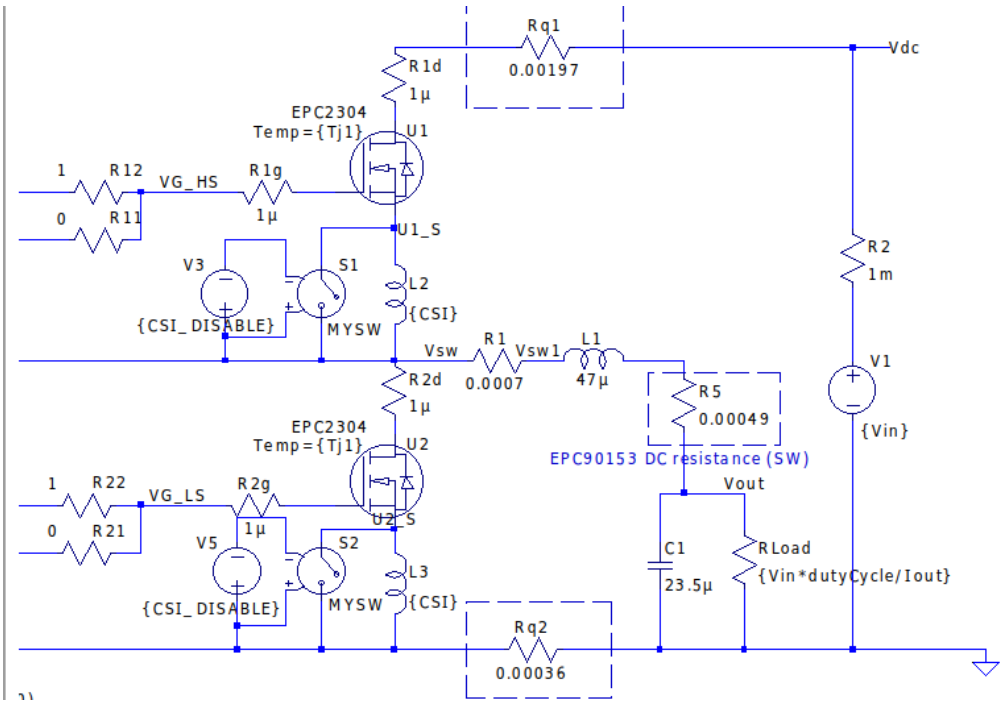

Figure 1.21: Schematic for switching behaviour

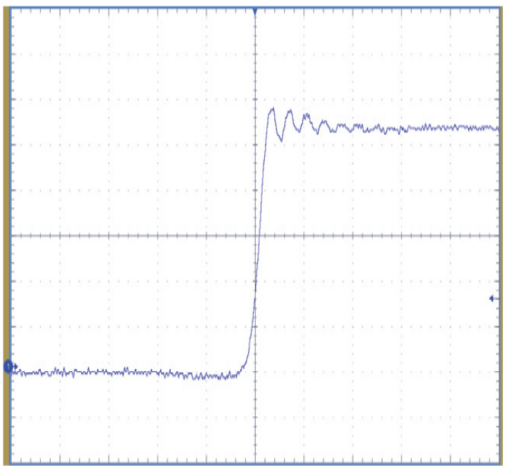

Figure 1.22: Datasheet Turn on curve

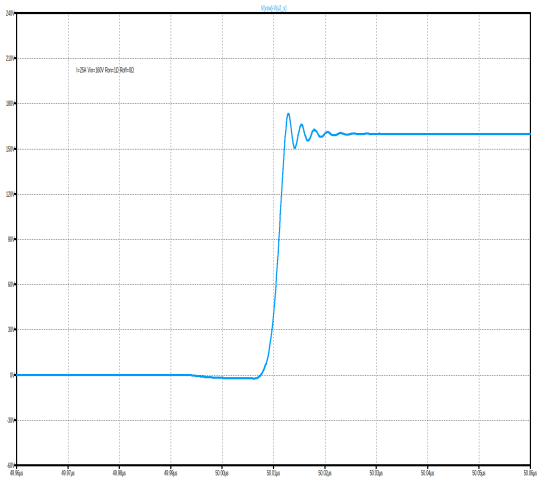

Figure 1.23: SPICE Turn on curve

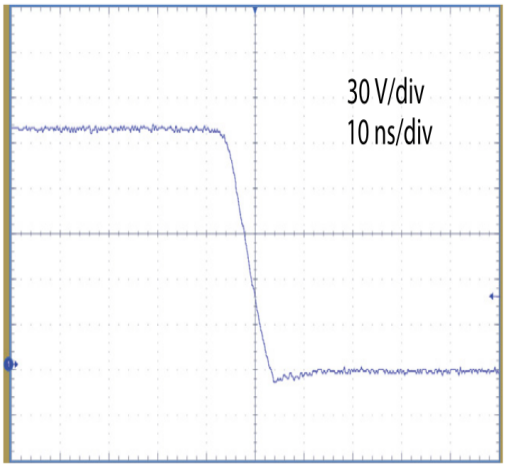

Figure 1.24: Datasheet Turn off curve

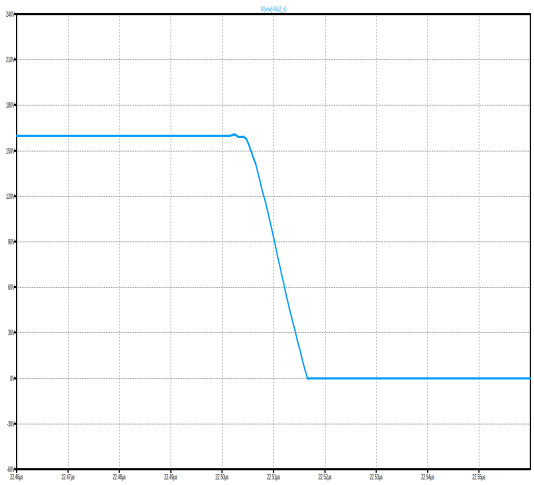

Figure 1.25: SPICE Turn off curve

Typical gate charge

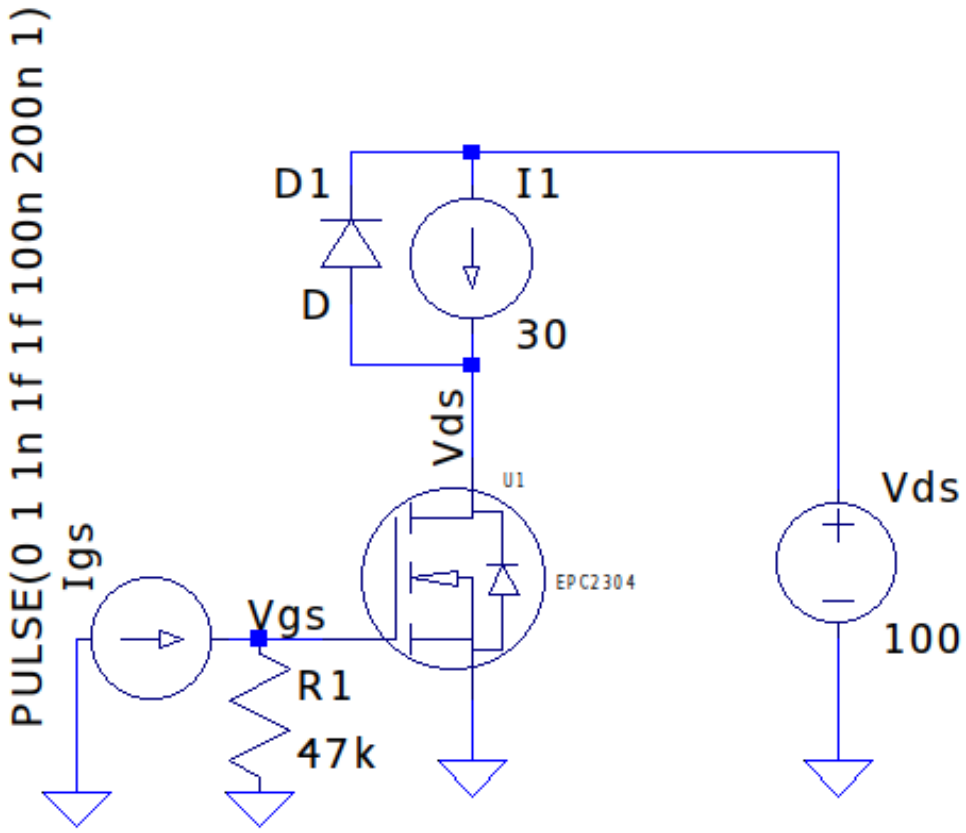

Figure 1.26: Schematic for gate charge

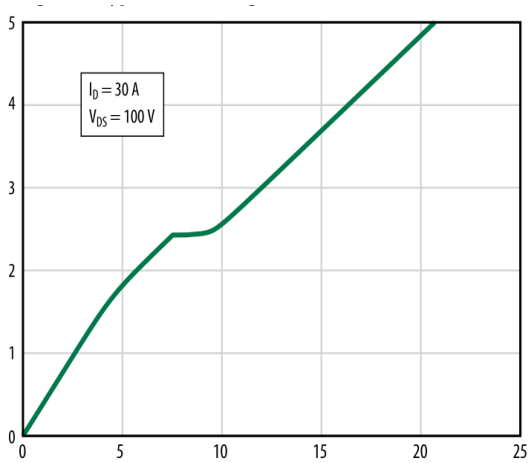

Figure 1.27: Datasheet gate charge

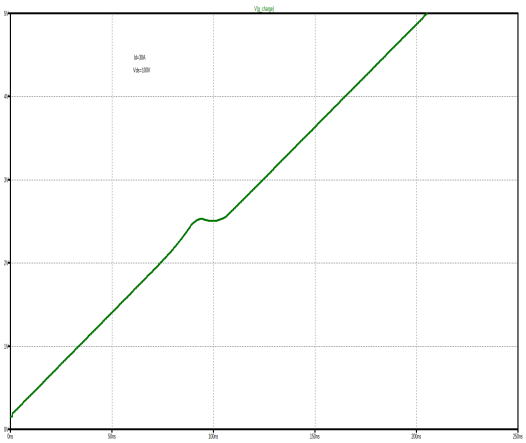

Figure 1.28: SPICE gate charge

Q (datasheet)	Q (simulation)
Qgtot=21nC	Qgtot=21.1nC
Qgs=7.5nC	Qgs=9nC
Qgd=2nC	Qgd=1.6nC
QVth=5.2nC	QVth=7.58nC

Table 1.6: Gate charge data comparision

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Bibliography

[1] Giorgio Armignacco. *Thesis repository*. https://github.com/arminioboy/PoliTOThesis. 2026 (cit. on p. 3).

[2] EPC-CO. *eGaN® FET DATASHEET EPC2304*. https://epc-co.com/epc/Portals/0/epc/documents/datasheets/EPC2304_datasheet.pdf. 2025 (cit. on p. 3).